

Predicting emergency medical service call demand: A modern spatiotemporal machine learning approach

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ABSTRACT

The primary goal of emergency medical service (EMS) agencies is to effectively allocate the ambulances and personnel required to provide sufficient geographic coverage of a service area while minimizing response times to high-priority call requests. Given that the demand for ambulances is known to fluctuate spatially and temporally based on the time of day and day of the week, EMS practitioners depend on call volume forecasts to develop staffing and dynamic redeployment plans. In this study, a series of daily, hourly, and spatially distributed hourly call volume predictions are generated using a multi-layer perceptron (MLP) artificial neural network model following feature selection using an ensemble-based decision tree model. For spatially distributed predictions, K-Means clustering is applied to produce heterogeneous spatial clusters based on call location and associated call volume densities. The predictive performance of the MLP model is benchmarked against both a selection of traditional time-series forecasting techniques and a common industry method. Results show that MLP models outperform time-series and industry forecasting methods, specifically at finer levels of spatial granularity where the need for more accurate call volumes forecasts is more essential.

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1. Introduction & research background

Emergency medical services (EMS), commonly referred to as ambulance, paramedic or pre-hospital emergency services, are a critical component in the delivery of urgent medical care to communities. The primary goal of any EMS agency is to minimize their response times to high priority emergency call requests and lessen the rate of mortality and morbidity [1]. EMS agencies are confronted with the evolving task of effectively allocating the ambulances and medical personnel required to provide sufficient geographic coverage while minimizing emergency response times, especially for high priority calls. To achieve these goals, EMS managers and dispatchers vigorously study the distribution of incoming call requests (demand) and establish resource deployment plans specifying the number of ambulances and emergency response personnel required for future periods. These deployment plans, derived from historical demand data and forecasts, determine the daily and hourly personnel work shift schedules [2,3]. Historically, the majority of studies have focused on time of the day as the sole dimension and have generated forecasts for broad geographic areas such as entire

counties or cities. However, the demand for ambulances is known to fluctuate temporally [4] and spatially based on the time of day and day of the week [5,6]. This requires dispatchers to redistribute (redeploy) their fleet of ambulances to different locations, often referred to as posts, throughout the day to compensate for spatiotemporal demand fluctuations. Industry and academic researchers have conducted numerous studies focused on developing novel deployment strategies, and associated staffing plans, in an effort to reduce response time while maximizing service coverage. These redeployment models are ultimately dependent on actionable call volume forecasts distributed by space and time to serve as inputs [1,7–10].

As noted by Aringhieri et al. [1], Channouf et al. [4], and McConnel and Wilson [11], some of the earliest works related to emergency medical services demand forecasting appeared in the 1970s [12–16]. Kamenetzky et al. [17] utilized a combination of census data for 1970, population and employment estimates for 1979, and complete sets of 1979 call data provided by 82 EMS agencies throughout 200 civil divisions in Southwestern Pennsylvania. Their final model included four of the 27 independent variables tested (population, employment, logarithm of the percentage of the population both white and married, the square of housing units per area resident), explaining 92% of the variations in demand for the areas where the total employment

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data were available. Baker and Fitzpatrick [18] applied a version of the Holt–Winters' exponential smoothing method, along with a parameter optimization model, to generate daily ambulance demand forecasts using data from four diverse South Carolina counties. Tandberg et al. [19] explored the performance of four time series models for generating hourly ambulance forecasts using 1994 call data provided by the Albuquerque Ambulance Service. Brown et al. [20] completed a study to measure the accuracy of forecasts generated using a popular demand pattern analysis method initially introduced as part of the System Status Management (SSM) framework [21]. Using 73 weeks of hourly call volume data collected from seven different EMS agencies, the researchers found that the SSM demand forecasting approach performed well with an approximate 19% estimation accuracy.

Channouf et al. [4] performed a study concentrated on generating daily and hourly EMS call volume forecasts by applying a variety of time-series methods. Emphasizing the importance of accurately predicting demand, Channouf et al. highlighted the fact that reliable forecasts serve as essential inputs into EMS planning models and accompanying staffing plans. The data used for their investigation was provided by the Calgary EMS System in Alberta, Canada and spanned a time-period of 50 months from 2000–2004. The researchers applied (1) a standard regression model, (2) a regression model configured with correlated residuals and (3) a doubly-seasonal ARIMA model to produce daily call volume forecasts and (1) a model “built around the conditional distribution of hourly volumes, conditional on the daily volume” and (2) a model “that fits a time-series directly to the hourly data” to generate hourly call volume forecasts. The authors found that each of the models generated relatively accurate forecasts at daily and hourly time scales.

Only in recent years have researchers recognized the importance of incorporating the temporal and spatial distribution of demand into their prediction models. In 2009, Setzler et al. [6] carried out an important study aimed at producing EMS call volume forecasts at various spatiotemporal granularities using artificial neural networks (ANN). The data used for their investigation consisted of emergency call responses recorded from 2002 to 2004 in Mecklenburg County, North Carolina. The authors aggregated the call data into 1-hour and 3-hour time-frame intervals and geographically into 2-mile and 4-mile fixed square-mile grids. It is important to note that the researchers selected these geographic granularity levels arbitrarily, and not based on any industry recommendations. They compared the call volume forecast values produced by their ANN against forecasts produced using an industry method commonly employed by EMS agencies, a unique adaptation of a 20-point moving average. Their results showed that the ANN approach moderately outperformed the moving average forecasts at the 4 x 4 mile 1-hour and 3-hour granularity levels. However, at the finer spatiotemporal granularities they noted that ANN did not produce more accurate forecasts. For 2 x 2 mile, 3-hour granularity there was no statistically significant difference between the ANN and moving average forecasts. At the 2 x 2 mile 1-hour scale the moving average forecasts were more accurate on average over the ANN approach. They discovered that at the finest spatiotemporal granularity over 90% of call volumes were zero, and in fact forecasting all zeros significantly outperformed both the industry method and the ANN. Consequently, they suggested that future studies determine near-optimal grid sizes and time-intervals.

Chen et al. [5] completed a complementary investigation comparing the performance of ANN, moving average, linear regression, and support vector regression to predict daily and sub-daily EMS call volumes for three districts within New Taipei City, Taiwan. Citing the works of Setzler et al. [6], the researchers further emphasized the importance of incorporating the spatial component of demand for producing accurate forecasts and

effectively allocating EMS resources. In assessing their results, the authors stated that their models produced acceptable daily call volume forecast accuracy with mean absolute percent error (MAPE) values ranging between 23.01% and 60.56% with an average of 40.66%. The forecast error values for sub-daily (3-hour time bucket) call volume estimations were much higher, with significantly lower overall accuracy.

Vile et al. [22] completed a study in 2012 in which they produced ambulance demand predictions using a novel time series forecasting technique known as Singular Spectrum Analysis (SSA). Referencing previous studies that applied time series methods, Vile et al. hypothesized that SSA could provide more accurate forecasts of EMS call demand. After collecting several years of ambulance call data provided by the Welsh Ambulance Service Trust in Wales, the researchers generated daily call volume forecasts using a combination of SSA, ARIMA and a Holts–Winters (HW) exponential smoothing models. The researchers found that SSA was capable of producing short-term forecasts (7–14 days) with accuracies equivalent to ARIMA and HW. Additionally, SSA generated forecasts with greater accuracy over longer periods of time (21–24 days). Vile et al. [22] state that in practice SSA is not only a comparable, or in some cases superior, forecast methodology, they argue that it is overall a more flexible approach capable of recognizing cycles and patterns in demand.

In this paper, a forecasting methodology that utilizes machine learning methods is proposed for producing, daily, hourly, and spatiotemporal call volume estimations at a degree of granularity in space and time that is practical and actionable. The study utilizes eight consecutive years of EMS call data (2010–2017). High priority call volume predictions are generated and compared using MLP, traditional time-series forecasting techniques, and a prominent industry model currently used by the participating EMS agency to generate hourly forecasts which we refer to in this paper as the MEDIC Hourly Forecasting (MHF) method. The MHF method is a unique adaptation of a traditional moving average formula that averages a total of 20 periods to produce estimations. The forecasted value for a given hour is calculated by averaging call volumes from the same hour of the day for the previous four days of the week time periods over the past five years. Citing the formula as a common industry approach, the MHF method was also used by Setzler et al. [6] and Chen et al. [5] as a benchmark method throughout their study. A definition of the MHF method formula is given in [Appendix C](#).

Input feature selection is conducted using an ensemble-based decision tree model and a novel method for spatial segmentation using k-means clustering is proposed as an alternative to fixed-grid geographic segmentation. The results show that MLP models consistently outperform the industry model and traditional time-series techniques when producing daily and spatiotemporally distributed call volume forecasts. The results of this study are expected to improve upon current industry practices and progress the current state of the art presented in the literature.

The remainder of the paper is organized as follows. Section 2 provides the data extraction and preprocessing details. Section 3 presents non-spatially distributed daily and hourly forecast methods and comparative results. Section 4 details the development and results of spatiotemporal forecasting methods. Conclusions and suggestions for future studies are summarized in Section 5.

2. Data extraction and pre-processing

The data used in this study was provided by an EMS Agency solely responsible for serving a large metropolitan (county) area located in the United States with an approximate population of one million. The original dataset contained 907,883 records, representing every 911 call EMS responded to over an eight-year period (January 1st, 2010 to December 31st, 2017). Each

call record contains the call date/time, call location coordinates, responding ambulance identifier, assigned call priority, patient problem description, call response outcome (transferred, refused care, false call, etc.), additional time stamps such as the time in route and arrival time, and a master incident number. Emergency medical calls (or incidents) are uniquely identified using a field labeled “master incident number”. While there is a one-to-one relationship between an individual call received by EMS and a single master incident number, the same master incident number may be logged for multiple records. This occurs in cases where multiple ambulances respond to a single call. For the purposes of this study, a total of 46,804 duplicate master incident number records were removed from the dataset, leaving 861,079 unique call records. The total annual call volume, shown in [Appendix A Fig. A.1](#), has increased each year with an average increase of 4.97% over eight years. Similar to the study conducted by Channouf et al. [4] this trend can be attributed to the steady population increase, from 923,326 in 2010 to 1,076,837 in 2017 (an avg. increase of 2.22%) and other factors such as aging of the citizens in the county.

To produce accurate short-term spatiotemporal call volume forecasts the call date, time, location, and call priority are all critically important. As previously noted, the primary goal of EMS agencies is to minimize their response times to high priority calls. For the purposes of short-term (daily and hourly) demand planning, the participating EMS agency indicated that call volume forecasts should focus exclusively on the highest priority one and two (P1+P2) calls. P1+P2 calls represent the most critical patients, require the fastest response times, and constitute the majority of all calls received (73.57%). Calls that do not require an immediate response are designated as non-emergent transports (NET), such as requests to transfer patients to and from clinical care facilities. A breakdown of the dispatch priority percentages for all calls recorded are shown in [Appendix A Table A.1](#). Given the objective of concentrating solely on producing high priority call predictions, 227,555 records representing all P3, NET, and other dispatch priority calls were dropped from the dataset. An additional 110 records that contained missing or incomplete fields were also dropped, leaving a total of 633,417 records in the final dataset.

To conduct a detailed analysis and pre-process the dataset for input into various forecasting models, a custom Python script and coding structure similar to Chen et al. [5] and Setzler et al. [6] were used to extract and label the year, season, month, week, day, day of the week and hour from call response date-timestamps. Additionally, daily and hourly call volumes were aggregated into two separate datasets by counting the number of call instances that occurred on individual days and within individual hours. In the hourly call volume data, a total of 199 instances of hours with no calls (i.e., zero call volume) were identified. To ensure a complete and accurate time series, zero call volume records for the missing 199-hour instances were generated using a separate Python script. At the daily scale, a seasonal component is present for the days of the week as illustrated in [Appendix A Fig. A.1](#). Sunday is consistently the one day of the week with the lowest total call volume each year, while Saturday typically experiences the second lowest call frequency. Furthermore, totals are consistently higher during the week (Monday–Friday) reaching peak demand on either Monday, Tuesday or Friday. Anecdotally, this can be attributed to increased activity throughout the county as people travel to and from work, school, and carry out routine weekly activities.

Drilling down to the hourly level, the distinction between weekdays (Monday–Friday) and weekend days (Sunday/Saturday) is more pronounced. [Fig. 1](#) shows the average hourly call volume by the day of the week for all 2010–2017 call data. A clear

oscillation pattern is visible at the hourly scale as call volumes decline in the early morning hours of the day to their lowest point at approximately 06:00. Call volumes increase significantly between the hours of 06:00 and 12:00, remaining at near peak levels for eight to ten hours each day. While call volumes are overall consistently higher during the weekdays, volumes remain closer to their daily peak volume levels for more hours during the afternoons of the weekend days compared to the weekdays. The patterns and trends present in the dataset are consistent with the behavior of a smaller dataset collected in Calgary, Alberta and used in the study conducted by Channouf et al. [4]. Their dataset also exhibited a clear long-term positive linear trend, monthly and daily seasonality. Feature selection processes and call volume aggregation schemes are discussed further in Sections 3 and 4.

3. Daily and hourly forecasts

Between 2010 to 2017, the participating EMS agency received an average of 217 calls per day and approximately 9 calls per hour. When scaled down to the daily and hourly level, EMS call volume levels tend to be stationary and repetitive in nature. As such, the use of traditional time series forecasting techniques such as moving average models, variations of exponential smoothing models, and autoregressive integrated moving average (ARIMA) models have historically been capable of producing daily call volume predictions for broad geographic areas with reasonable accuracy [4,19,23]. In this study, implementations of ARIMA, Holt-Winter's (HW) exponential smoothing method, and a multi-layer perceptron (MLP) model are used to produce daily call volume forecasts. For hourly call volume forecasts, the performance of a moving-average method employed by EMS practitioners (MHF method) is compared against ARIMA and an MLP model. To prepare the data for model processing, records were divided into training, validation and testing datasets. For all learning models (daily and hourly) the call volume records from 2010–2015 are used for training, while records from 2016 are used for validation, and 2017 records are reserved for testing. Following hyperparameter tuning and validation, the complete set of records from 2010–2016 are used for training the final models that generate 2017 test predictions.

3.1. Daily call volume forecasts

Prior to generating forecasts using $ARIMA_{(p,d,q)(p',d',q',m)}$, or some of the other traditional time-series techniques, it is important to first determine if a time series dataset is stationary. Stationarity is a fundamental pre-requisite for ARIMA to produce unbiased estimates. To evaluate the stationary nature of the time series, the Augmented Dickey-Fuller (ADF) test is applied to the 2010–2017 daily dataset [24]. The null hypothesis in this test is that the time series can be forecasted by a unit root (time-dependent structure). The alternate hypothesis is that the time series cannot be represented by a unit root and therefore is stationary. With a resulting p -value of 0.23, well above the 0.05 critical value threshold, the dataset is found to be non-stationary. Therefore, prior to deploying ARIMA, the sources of non-stationarity (i.e., trend in mean, variance, or covariance) need to be determined and the original non-stationary time series requires transformation into a stationary form. As illustrated in [Appendix A Fig. A.1](#), call volume experiences an upward trend during the 2010–2017 period. This underlying trend in mean is at least one source for non-stationarity. Hence, first-order differencing is applied to the time series and the ADF test is repeated. The test following differencing resulted in a practically zero p -value ($5.0E-29$), well below the 0.05 critical value threshold. This confirms that the source of non-stationarity was the trend in

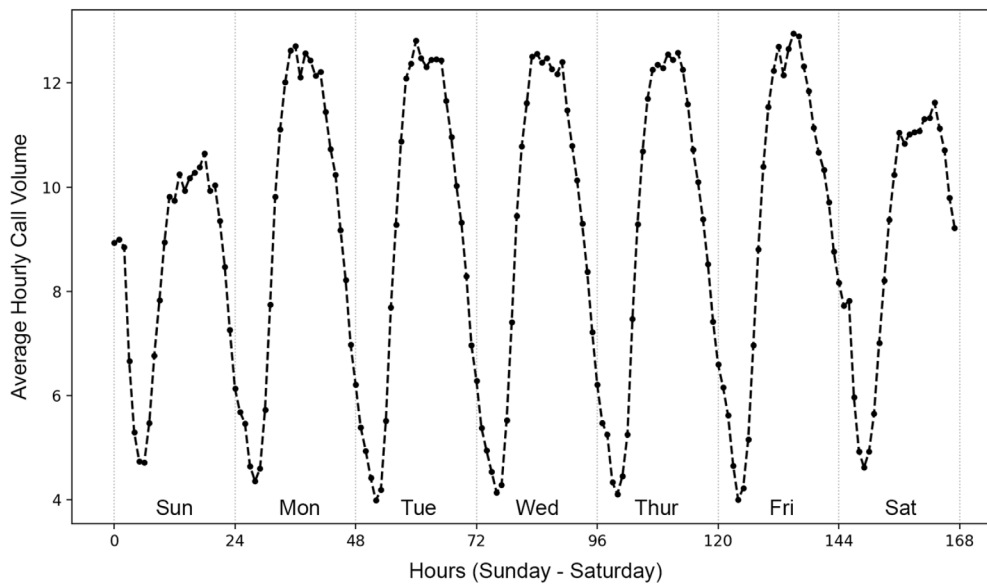


Fig. 1. Average P1+P2 call volume by day of week (2010–2017).

mean and that differencing is a sufficient step in transforming to a stationary time series. Since ARIMA carries out the differencing step innately, the original time series is used as input to ARIMA with the value of parameter d in $ARIMA_{(p,d,q)}(p',d',q',m)$ set to 1. To determine the optimal values for the parameters p and q , in the ARIMA model, an R implementation of the Auto-ARIMA function is utilized where Auto-ARIMA identified the optimal model configuration with the lowest Bayesian Information Criterion (BIC) value [25]. For daily forecasts, the seasonality parameter (m) is set to seven, representing the days of the week. Using the daily call volume records from 2010–2016 as the training dataset, Auto-ARIMA determined the optimal model configuration to be a seasonal $ARIMA_{(3,1,3)}(2,0,2,7)$.

Applying the HW exponential smoothing method, the initial seasonal index values were estimated by calculating the least squares trend line using the first two weeks of call volume actuals and then dividing those actuals by their corresponding trend line values. That resulted in fourteen values for each day of the two-week sample period. Those values were then averaged based on the day of the week, resulting in seven initial seasonal index values for each day of the week. The initial base level was then computed by dividing the seventh period value in the first seasonal cycle by its estimated seasonal index value. The long-term increase in the daily call volumes was then estimated using the slope of the first ten weeks (70 observations) which was 0.15, hence we set the initial trend value to 0.15. The smoothing parameter values (α , β , and γ) were tuned through optimization using the evolutionary search algorithm in Excel Solver. With the objective function set to minimize the MAPE value for 2017 predictions, and the seasonality configured to seven, the optimal parameter values were determined to be 0.0351, 0.0100, and 0.0286, for α , β , and γ , respectively.

In the two previous studies that explored ANNs as an approach for generating call volume estimations, both Setzler et al. and Chen et al. employed instances of MLP models. MLP is a supervised machine learning algorithm that uses a technique known as backpropagation to train an estimation model given a training dataset containing (x) number of input features [5,6]. The structure of an MLP network consists of a series of layers, each containing a collection of nodes (neurons). Represented mathematically as a directed graph, the individual nodes at each layer are fully connected to the nodes of the following layer.

Each network contains a single input layer, an output layer, and one or more hidden layers for processing inputs. An illustration of a generic MLP model with a single hidden layer is shown in Appendix B Fig. B.1.

3.1.1. Feature selection

For any predictive model to perform well, the proper input data (predictors/features) need to be identified. Feature selection, also referred to as attribute or variable selection, is the process of systematically selecting a subset of attributes from a dataset that are determined to be the most relevant to a prediction problem. Hence, feature selection serves as an important pre-processing step for developing effective learning models and removing noisy data.

In this study, a novel feature selection method known as Boruta is applied to identify all relevant features in EMS call volume datasets [26]. The literature [5,6] previously identified several temporal features (year, season, month, week, day, day of week) as candidate features for forecasting daily call volumes. Lastly, based on the structure of the seasonal $ARIMA_{(3,1,3)}(2,0,2,7)$ model, distributed lag values of periods 1, 2, 3, 7 and 14 for the daily features (day, day of the week, and volume) were added to the candidate feature set. The inclusion of the lag variables for call volume in the MLP model results in rolling window predictions which are comparable to time-series forecasting methods (e.g., ARIMA, exponential smoothing, moving average).

After identifying all the candidate features for the daily predictions, Boruta (implemented in R) was used to determine which final features should be included in the model [26]. Boruta functions as a wrapper method leveraging ensemble-based learning algorithms, specifically variations of decision-tree ensembles. For each candidate feature, Boruta creates a shadow feature by shuffling the values of the original feature. This shadow feature maintains the same mean and standard deviation of the original feature, yet it should not have any significant correlation with the original variable or target variable (as it has been created by a random process). After creating shadow features for each original feature, Boruta compares the predictive power of each original feature against their shadow features. If any of the original features have a lower predictive power compared to their corresponding shadow feature, the algorithm labels those features as “unimportant”. If the predictive power of an original feature is

higher than that of its shadow feature, the algorithm will label it as “important”. The final output from the Boruta algorithm is a list of “important” input features for use in subsequent learning models. After running Boruta using a random forest regressor, the feature day and its corresponding lag variables were rejected. The remaining candidate features were confirmed to be important for estimating daily EMS call volumes.

3.1.2. ANN development for daily forecasts

This study utilized the Python-based machine learning library scikit-learn to develop and implement MLP regression models for generating call volume predictions. One notable disadvantage of MLP models is that network weights are initialized randomly. As a result, performance levels naturally vary slightly from run to run. One strategy to overcome this issue is to cycle through numerous iterations during validation testing and select the model instance with the highest predictive performance for each model configuration. Therefore, a total of 1000 iterations were performed for each model configuration and the instance with the lowest MAPE across the validation dataset was recorded. Cross validation strategies, such as K-folds, that iteratively train models using slices of a given dataset ignore the temporal components inherent to time series data and consequently are not beneficial in this setting. As an expansion of earlier investigations, the initial MLP models tested were initialized based on the final parameter settings used by Setzler et al. and Chen et al. Each configured their MLP networks with the number of input nodes matching the number of features, a single hidden layer mirroring the number of nodes from the input layer, and a sole output node representing the volume estimation values. In reviewing best practices for forecasting with ANNs, Zhang et al. [27] evaluated historical findings and techniques for determining the number of hidden layers and nodes in an ANN. They concluded that, for the majority of forecasting problems, one to two hidden layers are appropriate for producing accurate results. When determining the optimal number of hidden nodes, Zhang et al. noted that networks with fewer hidden nodes are typically preferred as they produce more generalized models, reduce the likelihood of overfitting the model based on the training data, and have lower complexity.

To commence MLP model tuning a set of initial hyperparameters were explored through an exhaustive grid search, with the resulting MAPE value serving as the key performance metric. The logistic sigmoid, hyperbolic tangent, and rectified linear unit activation functions were evaluated along with the “lbfgs”, stochastic gradient descent, and stochastic gradient-based optimizer solver functions. Initial tests using the stochastic gradient descent and stochastic gradient-based optimizer solver function confirmed that an adaptive learning rate consistently produced better results. Lastly, the hidden node configuration strategy described by Zhang et al. was evaluated in the initial grid search using one to two hidden layers and hidden node sizes equal to 20, x , $x+1$, $x-1$, $2x$, 50, 100, 150, 200, and 300 where (x) represents the number of input features. Using the complete set of relevant features identified during feature selection, multiple rounds of random and grid search found models configured with the logistic sigmoid activation function, the stochastic gradient-based optimizer solver function, a single hidden layer, and 100 hidden nodes outperformed alternative configurations.

3.1.3. Daily call volume forecast results

To compare the estimation performance across the three daily forecasting methods, the MAPE values were calculated using the daily call volume records from 2017. Mean absolute deviation (MAD) values were also calculated for each method to provide perspective on the average number of calls under or over forecasted. All three methods resulted in accurate daily call volume

estimations, with the MLP model yielding the lowest MAPE of 5.98% with a MAD of 14.74. The HW's triple exponential smoothing method using optimal parameter values resulted in a MAPE of 6.22% and a MAD of 15.20, whereas ARIMA_{(3,1,3)(2,0,2,7)} performed a bit further behind with a MAPE of 7.31% and a MAD of 17.42. These findings suggest that EMS agencies can easily generate accurate daily call volume forecasts, which are used for work force and vehicle scheduling, using any of the three approaches compared in this study.

3.2. Hourly call volume forecasts

Hourly call volume forecasts serve as critical inputs into detailed work-force schedules (peak and off-peak hourly demand periods). To generate hourly forecasts, an MLP model is further developed based on the performance of daily forecasting models and results are compared against the MHF benchmark method currently used by the participating EMS agency for generating hourly call volume estimations. ARIMA and HW's triple exponential smoothing are also employed as additional models of comparison. To determine the stationary nature of the hourly time series, the ADF test was performed against the complete hourly call volume dataset, yielding a p -value of $4.1E-28$, verifying that the series is stationary. To produce hourly forecasts using ARIMA, the seasonality parameter (m) is set to twenty-four. Auto-ARIMA identified the optimal hourly model configuration to be ARIMA_{(1,1,1)(0,0,4,24)}. The optimal parameter values to generate hourly forecasts using the HW exponential smoothing method were determined using the same process summarized for daily forecasts. To address zero call volume instances, present at the hourly level, and ensure that MAPE values were calculable and comparable across all models and methods, a value of one was added to all of the original hourly call volume actuals prior to training.

3.2.1. Feature selection

Feature selection for hourly call volume MLP models was completed following the same process utilized for daily call volume models. In addition to the features previously identified (year, season, month, week, day, day of week, hour) lag values of hourly call volume for periods 1–7, 24, 48, 72, 96, 168 were added to the candidate feature set based on the optimal hourly ARIMA_{(1,1,1)(0,0,4,24)} model. Using a random forest regressor, Boruta confirmed all features as relevant for producing hourly EMS call volume estimations.

3.2.2. ANN development for hourly forecasts

Based on the performance of the MLP model developed for generating daily call volume predications, a subsequent MLP model was initialized with comparable configurations for producing hourly call volume predictions. As before, the call volume records from 2010–2016 serve as the training/validation dataset and 2017 records are used for testing. Consistent with the grid-search strategy applied to daily MLP models, hourly MLP models having (x) number of input features were evaluated using hidden node sizes equal to 20, x , $x+1$, $x-1$, $2x$, 50, 100, 150, 200, and 300 with one to two hidden layers. Multiple rounds of hyperparameter tuning were completed using the array of activation functions, solver functions, and hidden node/layer combinations. Through grid-search the MLP model yielding the lowest MAPE leveraged the hyperbolic tangent activation function, stochastic gradient descent solver function, and a single hidden layer containing 100 hidden nodes.

3.2.3. Hourly call volume forecast results

To compare the estimation performance across the hourly forecasting methods, MAPE values were calculated using the hourly call volume records from 2017. Testing against the complete annual dataset provides a comprehensive benchmark to compare the adaptive nature of a machine learning model for one-step ahead time-series forecasting, i.e., MLP versus rolling moving-average methods such as MHF. This strategy also ensures the seasonal adjustment (hourly, daily, weekly, monthly, etc.) capabilities of each method are thoroughly evaluated. ARIMA produced the worst results at the hourly scale with a MAPE of 41.37% and a MAD of 3.52. The HW method resulted in a MAPE of 33.96% and a MAD of 2.60, while the MHF method performed the best with a MAPE of 27.14% and a MAD of 2.67. The MLP model yielded the second lowest MAPE value of 27.31% and a MAD of 2.79.

4. Spatiotemporal forecasts

Accurate hourly call volume forecasts that include the spatial component of demand are critical to supporting real-time operational activities such as ambulance dispatch decisions and initiating dynamic redeployment strategies aimed at perpetually maximizing response coverage across a defined geographic service area. As previously noted, as calls are received, the latitude and longitude coordinates representing the originating call locations are recorded. This information is useful for conducting spatial demand analysis and identifying areas that regularly experience higher call volumes. To incorporate the spatial component of demand into temporal forecasting models, a K-Means clustering approach is implemented to separate calls into related groupings based on call location densities.

4.1. Spatial clustering

As discussed in the literature review, Setzler et al. [6] and Chen et al. [5] incorporated the spatial component of demand to generate demand estimations at various spatiotemporal granularities. In each investigation, the researchers developed fixed square-mile/kilometer grid block systems to segment geographic areas and formulate estimations. In doing so, they encountered a key limitation of this approach; zero-inflated demand distributions that are created when scaled to finer degrees of spatial granularity. After presenting their results, Setzler et al. stated that at a certain degree finer levels of specificity have little practical value and suggested future investigations focus on “varying population densities in order to determine optimal or near-optimal geographic grid sizes and time intervals”.

The relationship between population density and call volume densities has been recognized in numerous studies [5,10–14]. Employing even simple methods, such as linear regression, consistently reveals strong relations between call volumes and population counts. While fixed geographic grid layouts provide a straightforward approach to spatial segmentation, the equal division of space does not account for the unequal distribution of populations and associated call demand. Alternatively, a general-purpose clustering algorithm, such as K-Means, can be applied as a data mining technique to produce geographically heterogeneous spatial clusters based on call location (latitude and longitude coordinate) and associated call volume densities. K-Means is a clustering algorithm that divides a collection of N samples into K distinct clusters, each defined by the average (centroid) of the samples in a cluster. The objective of the K-Means algorithm is to minimize the within-cluster sum of squared (WCSS) error (Euclidean distance) between sample points and a cluster

centroid [28]. Since data points are grouped based on proximity, K-Means is a common approach for creating spatial clusters using latitude and longitude coordinates. In the prolific geographic information systems platform ArcGIS, K-Means is one of the primary unsupervised machine learning methods used by the ArcMap grouping analysis tool to identify natural groupings in spatial datasets. Moore and Dixon [29] utilized the grouping analysis tool to cluster instances of tornados produced by Hurricane Ivan in 2004 based on their latitude/longitude coordinates and relative time of touchdown. Naaman et al. [30] also explored a variation of K-Means for automatic organization of digital photos based on geographic coordinates and time-based event information.

Establishing clusters of varying sizes based on call location volume results in groupings that are more representative of associated population density and reduces the likelihood of generating zero-inflated distributions. As suggested by Setzler et al. a practical level of granularity is required to produce operationally valuable demand estimations and support real-time dispatching decisions. Citing average travel times, distances, and the size of the study area, operations managers at the participating EMS agency stated that approximately seven to nine spatial clusters would serve as an optimal level of granularity for utilizing hourly call volume forecasts for real-time redeployment decisions [31]. This equates to three different K-Means model configurations (i.e., $K = 7$, $K = 8$, and $K = 9$). To further evaluate the optimal number of clusters (i.e., the optimal level of granularity) a technique known as the elbow method is applied. The elbow method aids in identifying an optimal or near-optimal value for K by fitting the model with a range of values for K and calculating the average within-cluster sum of squared (WCSS) error for each cluster set. The WCSS error values for each model are then plotted on a line chart, which frequently resembles the shape of an arm if the data is properly clustered. The elbow, or inflection point on the curve, is an indicator of the optimal value, or range of values, of K . After applying the elbow method to our call location dataset, 5–10 clusters appear to be the optimal range based on the declining rate of change of the within clusters sum of squared error. This supports the anecdotal recommendation of the participating EMS agency that 7–9 spatial clusters serve as an optimal level of granularity for hourly call volume planning. The resulting elbow method chart is shown below in [Appendix D Fig. D.1](#). Using the original P1+P2 call records, the latitude and longitude coordinates for each call were extracted and processed through a K-Means clustering algorithm implemented in scikit-learn. Upon convergence, a numeric label identifying the corresponding spatial cluster for each call was appended to each of the original call records. This process was repeated for values of $K = 7, 8$, and 9 . The resulting K-Means clusters are visualized below in [Fig. 2](#). The call volume counts per cluster are listed in [Appendix D Table D.1](#).

4.2. Spatially distributed hourly call volume forecasts

As an extension of the non-spatial hourly call volume forecasts, the performance of an MLP model is compared against the MHF method, HW, and ARIMA for producing spatially distributed hourly call volume forecasts. In preparation for model processing, the hourly call volumes for the range of spatial cluster sets ($K = 7, 8, 9$) were aggregated separately by counting the number of call instances that occurred within each cluster during individual hours. This resulted in three spatially distributed hourly call volume datasets for 7, 8 and 9 spatial clusters respectively. To ensure a complete and accurate time series, zero call volume records for any missing cluster/hour instances were generated and inserted into to each dataset. As before, a value of one was added to all of the spatially distributed hourly call volume

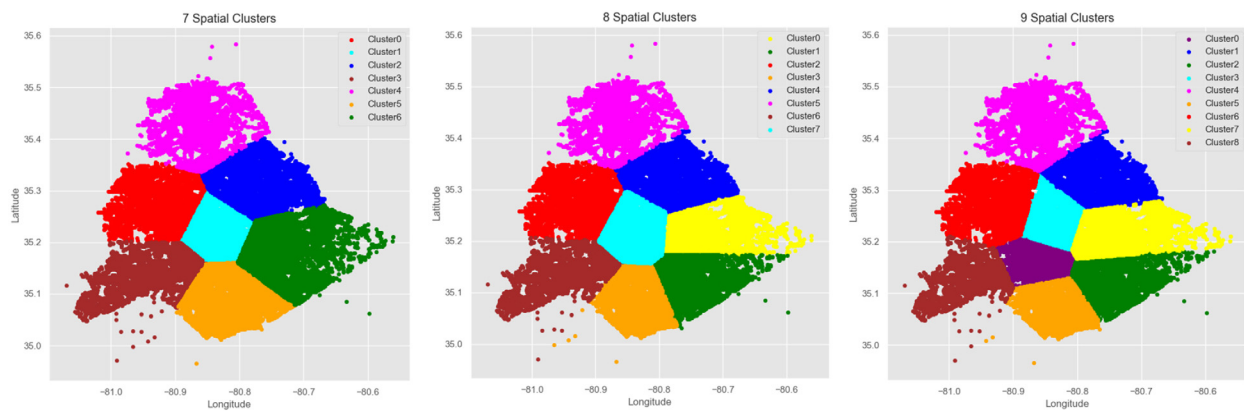


Fig. 2. P1+P2 calls K-means spatial clusters; K=7, 8, 9 (2010–2017).

actuals prior to training to address zero call volume instances and ensure that MAPE values were calculable and comparable across all models and methods. The ADF test was also conducted again to verify that each of the newly generated time series datasets are stationary prior to model processing. Processing each of the cluster datasets individually, spatially distributed call volume estimations using the optimal hourly $ARIMA_{(1,1,1)(0,0,4,24)}$ model and optimized HW exponential smoothing model were produced. The optimal HW exponential smoothing parameter values to generate spatially distributed hourly forecasts were identified using the process previously described for daily and hourly forecasts. For the MHF benchmark method, call volume prediction values for individual hours were generated by iteratively processing each cluster dataset separately using the corresponding day of the week, hour, and k-means spatial cluster label. This resulted in spatially disturbed hourly call volume forecasts by cluster using ARIMA, HW and the MHF method.

4.2.1. Feature selection

Feature selection for spatially distributed hourly call volume MLP models was carried out following the same process employed for hourly and daily call volume models. With the addition of the K-Means clustering label for each of the three spatial cluster sets ($K = 7, 8, 9$) the complete candidate feature set included: year, season, month, week, day, day of week, hour, the K-Means cluster label, and lag values of hourly call volume per cluster for periods 1–7, 24, 48, 72, 96, 168. Processing each of the three datasets separately through a random forest regressor, Boruta consistently confirmed all features as relevant for estimating spatially distributed hourly EMS call volumes.

4.2.2. ANN development for spatially distributed hourly forecasts

MLP model development, hyperparameter tuning, and implementation for generating spatially distributed hourly call volume estimations was carried out following the same process used for producing hourly non-spatial call volume predictions. Each model instance was trained using the input features and hourly call volume target values from each dataset for all clusters, with the K-Means cluster label serving as the spatial identifier in each configuration $K=7, 8, 9$. Training the MLP instances using the data for all clusters allowed the models to adapt not only to the patterns and trends of an individual cluster's call volume behavior, but also to the patterns and trends of analogous clusters. Following multiple rounds of hyperparameter tuning, the top performing MLP model was a deep neural network configured with the hyperbolic tangent activation function, the stochastic gradient descent solver function, and two hidden layers each containing 200 hidden nodes.

4.2.3. Spatially distributed hourly call volume forecasts results

To compare the estimation performance across the MHF method, HW, ARIMA, and MLP models, MAPE values were calculated separately for each cluster using 2017 records from each of the three spatially distributed hourly call volume datasets ($K = 7, 8, 9$). Consistent with the non-spatially distributed hourly forecasts, ARIMA performed the worst with average MAPE values of 53.48%, 54.93%, and 51.55% for spatial cluster sets 7, 8, and 9 correspondingly. HW yielded average MAPE values of 50.45% with 7 clusters, 49.36% with 8 clusters, and 47.84% with 9 clusters. The performance of the MHF method improved slightly as the level of cluster granularity increased with average MAPE values of 48.57% at 7 clusters, 47.59% at 8 clusters, and 46.13% at 9 clusters. The MLP models considerably outperformed the other methods with average MAPE values of 39.45%, 38.11%, and 35.56% across 7, 8, and 9 cluster sets respectively. Results for each method at the per cluster level using 9 clusters are outlined in Table 1.

4.3. Discussion

When comparing the performance of spatially distributed hourly forecasts generated using the MHF benchmark method, the MLP model consistently yields lower estimation error across all spatial clusters in terms of MAPE. Additionally, the MLP model predictions are found to be more accurate for clusters containing relatively lower call volumes (clusters with 10% or less of the total county-wide call volume). Using the 9-cluster configuration for visual analysis, Appendix E Fig. E.1 presents the MAPE values and percent differences between the MLP and MHF methods on a choropleth map shaded based on the percent of total call volume by cluster. The distinction in predictive performance levels between the MLP and MHF methods at the per cluster level is also apparent in terms of absolute deviation (error) when examining the error distribution per cluster (refer to Appendix E Fig. E.2) (see Fig. E.3).

Both approaches are shown to produce higher MAPE values in clusters containing higher (>10%) call volume (and associated population) densities (i.e., clusters 1, 3, 6, and 7). Geographically, these clusters represent urban centers within the county and include a greater concentration of mixed-use developments, multi-family housing units, and commercial buildings. These higher density clusters also encompass, or are adjacent to clusters containing, several hospitals with emergency departments. According to practitioners at the participating EMS agency, at any given time there are multiple ambulances located in and around the hospitals either transferring patients to the facility or completing paperwork. Hence, a major challenge of deployment planning is providing sufficient ambulance coverage across a mix of urban, suburban, and rural areas. Suburban and rural areas

Table 1
Per cluster MAPE/MAD results by method.

Cluster #	ARIMA MAPE	ARIMA MAD	HW MAPE	HW MAD	MHF MAPE	MHF MAD	MLP MAPE	MLP MAD
0	48.41	0.79	49.62	0.82	48.03	0.81	37.15	0.79
1	44.88	0.88	51.36	0.91	48.14	0.90	38.75	0.94
2	54.57	0.74	45.89	0.70	44.79	0.71	32.70	0.67
3	51.89	1.33	51.97	1.25	49.65	1.26	45.25	1.21
4	58.66	0.72	43.79	0.65	41.56	0.65	28.72	0.62
5	55.55	0.73	45.02	0.68	43.78	0.69	30.66	0.68
6	45.69	0.87	52.46	0.91	51.35	0.91	41.98	0.96
7	63.75	1.14	51.84	1.04	52.14	1.05	43.63	1.09
8	40.52	0.58	38.58	0.55	35.71	0.53	21.16	0.46
Average =	51.55	0.86	47.84	0.84	46.13	0.83	35.56	0.82

are exceptionally difficult for coverage planning; given lower call frequency and increased travel distances [31]. Therefore, a forecasting model, such as MLP, that produces more accurate spatiotemporal call volume estimations than the current industry method (MHF) is critical to providing adequate coverage and minimizing call response times. Particularly in lower call volume areas where the need for accurate predictions is greater. The MLP method was also found to produce similar results with the inclusion of explanatory weather data variables (refer to [Appendix F](#) for expanded details and results).

5. Conclusions and suggestions for future research

The objective of this study was to present a forecasting methodology that utilizes machine learning methods to generate daily, hourly, and spatiotemporal call volume estimations at a degree of granularity in space and time that is both practical and actionable. Forecasting model results demonstrate that, given systematic feature selection and hyperparameter model tuning, MLP models consistently produce more accurate predictions. For non-spatially distributed daily and hourly call volume predictions, traditional time-series and the MHF benchmark methods are shown to perform at marginally similar levels of predictive performance without the added complexity of machine learning methods. These results are consistent with the findings of previous studies by Setzler et al. [6] and Chen et al. [5]. However, when producing spatially distributed call volume forecasts, MLP models significantly outperform traditional time series methods across all spatial levels evaluated (7, 8, and 9 clusters). This finding has practical implications given the need practitioners have for accurate call volume predictions at more granular spatiotemporal levels.

Future studies have the potential to further advance the performance of spatiotemporal call volume prediction models by incorporating explanatory variables related to population shifts and reliable weather forecast data (i.e., temperatures, rainfall, etc.) at the hourly level. Datasets related to traffic patterns and population densities distributed by time and space collected through services such as Google Maps™ and Waze™ could be introduced to future models. It is important to note that the optimal level of spatial granularity will vary based on the service area size, geographical and topographical features, and population densities. The effectiveness of alternative clustering methods should be explored, and EMS practitioners should be consulted to aid in identifying practical levels of spatial granularity for use in subsequent investigations. To overcome the challenges associated with evaluating various forecasting methods given zero-inflated demand distributions at finer degrees of spatial granularity, future researchers could also explore using alternative metrics of predictive performance such as those discussed by Hyndman and Koehler [32]. Lastly, researchers could compare the effectiveness

Table A.1
Dispatch priority call percentages.

Dispatch priority	Description	# of calls	% of calls
P1	Life Threatening	160,912	18.69%
P2	Potentially Life Threatening	472,615	54.89%
P3	Non-Life Threatening	90,561	10.52%
NET/Other	Non-Emergent transport	136,994	15.91%

of the applications of the MLP and alternative machine learning models with the SSA approach [22] for generating EMS call volume predictions.

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Appendix A. Data extraction and pre-processing

See [Fig. A.1](#) and [Table A.1](#).

Appendix B. MLP model structure

See [Fig. B.1](#).

Appendix C. MEDIC Hourly Forecasting (MHF) method

The MEDIC Hourly Forecasting (MHF) Method equation is defined as:

$$F_{h,d,w,y} = \frac{\sum_{i=0}^4 \sum_{j=1}^4 A_{h,d,w-j,y-i}}{20} \quad (1)$$

where the following notation applies:

- F = forecasted call volume
- A = actual call volume
- h = hour
- d = day of the week (Mon, Tue, Wed, Thu, Fri, Sat, Sun)
- w = week
- y = year

Appendix D. Spatial clustering

See [Fig. D.1](#) and [Table D.1](#).

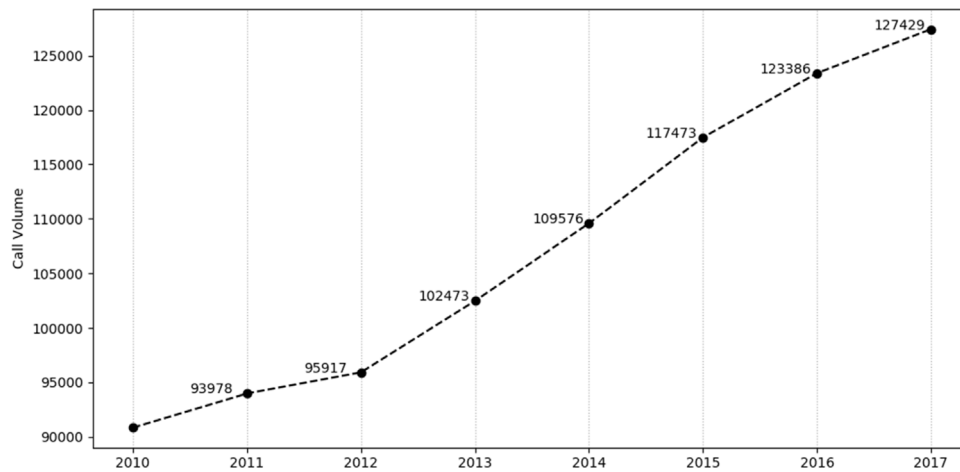


Fig. A.1. Total call volume by year.

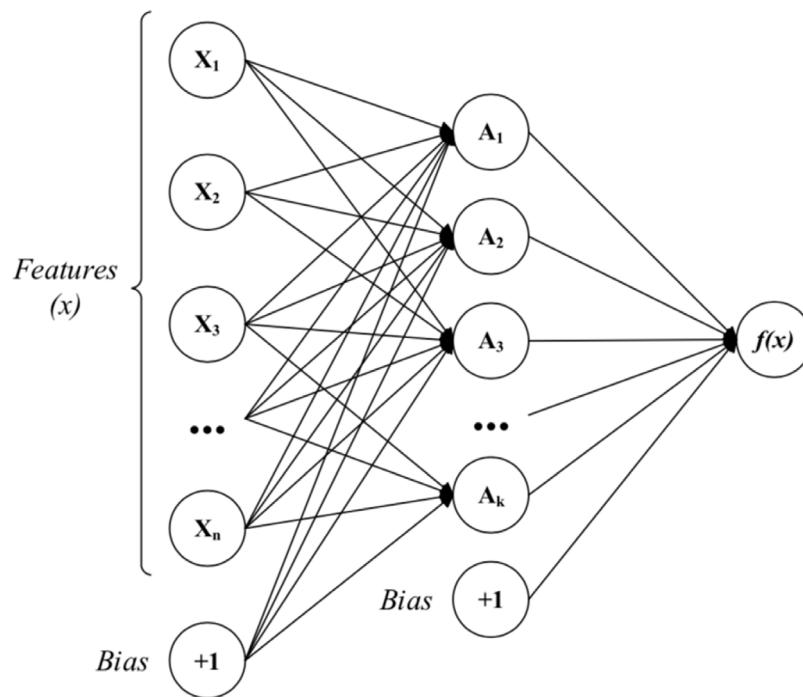


Fig. B.1. Single hidden layer MLP model.

Table D.1

Total call volume counts per cluster.

Total call volume per cluster						
Cluster #	K = 7	% of total	K = 8	% of total	K = 9	% of total
0	95416	15%	102252	16%	65065	10%
1	164594	26%	50723	8%	74044	12%
2	89018	14%	87136	14%	49174	8%
3	58100	9%	62441	10%	147307	23%
4	37516	6%	82883	13%	37062	6%
5	72756	11%	34282	5%	45524	7%
6	116017	18%	51004	8%	77317	12%
7	-	-	162696	26%	113314	18%
8	-	-	-	-	24610	4%

Appendix E. Spatially distributed hourly call volume forecasts results

See Figs. E.1–E.3.

Appendix F. Expanded MLP model with lagged weather variables results

Several EMS researchers have considered the inclusion of explanatory weather variables in their demand forecast models in an effort to improve prediction accuracy. Wong and Lai [23] and McLay et al. [33] performed novel studies analyzing the effects of weather on historical ambulance services demand. In 2014, Wong and Lai [34] carried out a follow-up study focused entirely on developing short-range forecasts using weather factors as causal predictors. To evaluate the addition of explanatory weather variables in this study, weather data was collected from the National Oceanic and Atmospheric Administration (NOAA). The dataset included the historic average daily temperature, minimum daily temperature, maximum daily temperature, precipitation in inches, and binary variables indicating occurrences of daily rain and snow fall for the geographic study area. Since this weather data represents historical actuals and not recorded

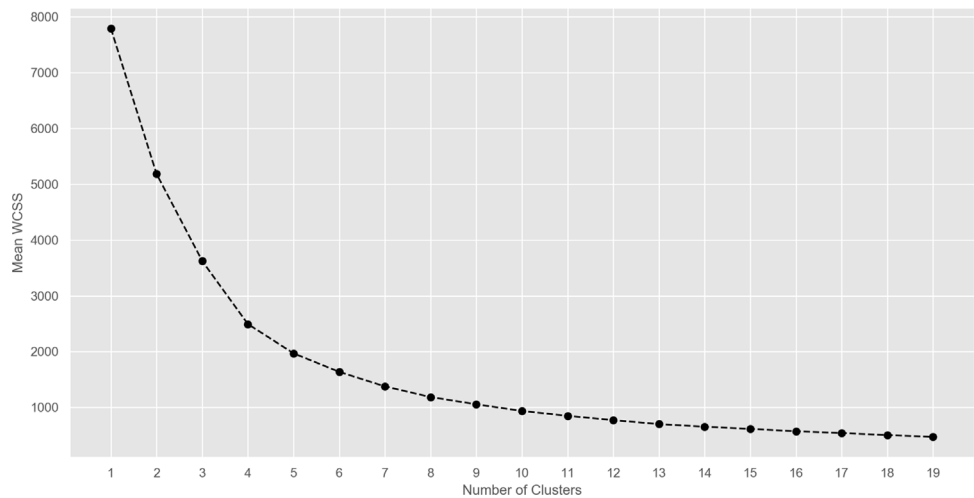


Fig. D.1. P1+P2 spatial clusters elbow method chart.

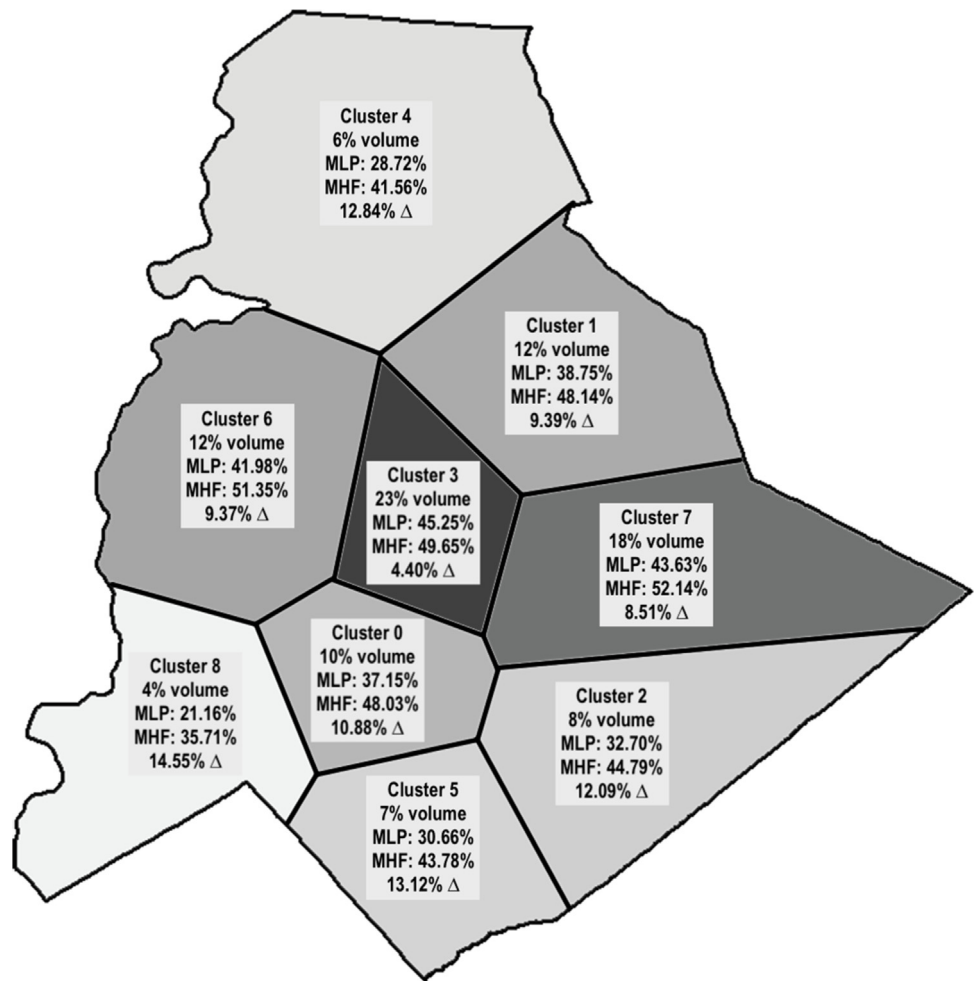


Fig. E.1. Percent of total call volume and MAPE values (MLP vs. MHF) by cluster. Note: Fig. E.1 presents a choropleth map shaded in grayscale based on the percent of total call volume by cluster. For each cluster the percent of total call volume is provided along with the MAPE values for MLP, MHF, and the MAPE improvement delta that MLP resulted in over MHF.

weather forecasts, lagged versions of these weather variables were created and added to the feature sets following the same process of feature selection and training previously employed for daily, hourly, and spatiotemporal models. Subsequent model results (presented in Table F.1) show that the inclusion of lagged weather variables does not improve prediction accuracy. These

results are consistent with the findings of Wong and Lai [34], who cited that due to the fact that not all EMS demand is related to weather the inclusion of weather variables may, in some cases, reduce or have no positive impact on forecast accuracy.

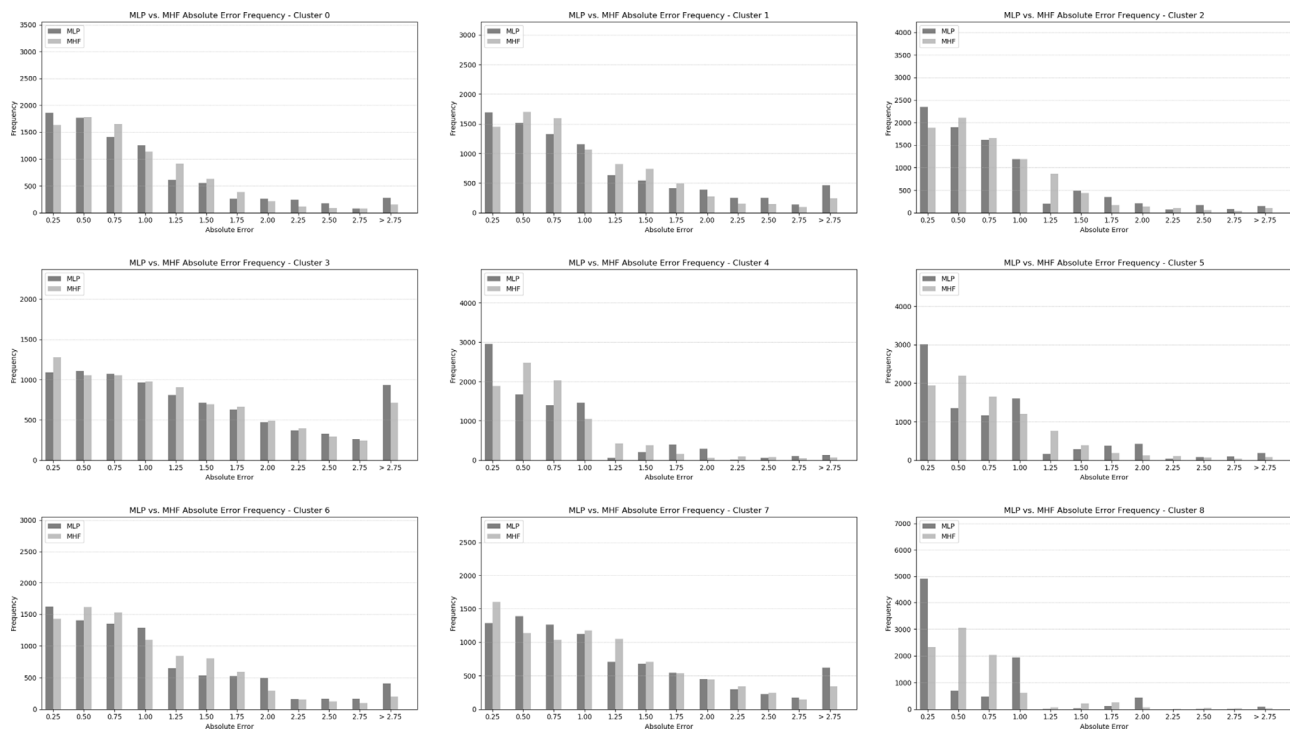


Fig. E.2. MLP vs. MHF absolute error frequency distributions by cluster. Note: Fig. E.2 contains a series of tables displaying the absolute error frequency distribution for the MLP and MHF methods by cluster. MLP results in lower absolute estimation errors more frequently for lower volume clusters.

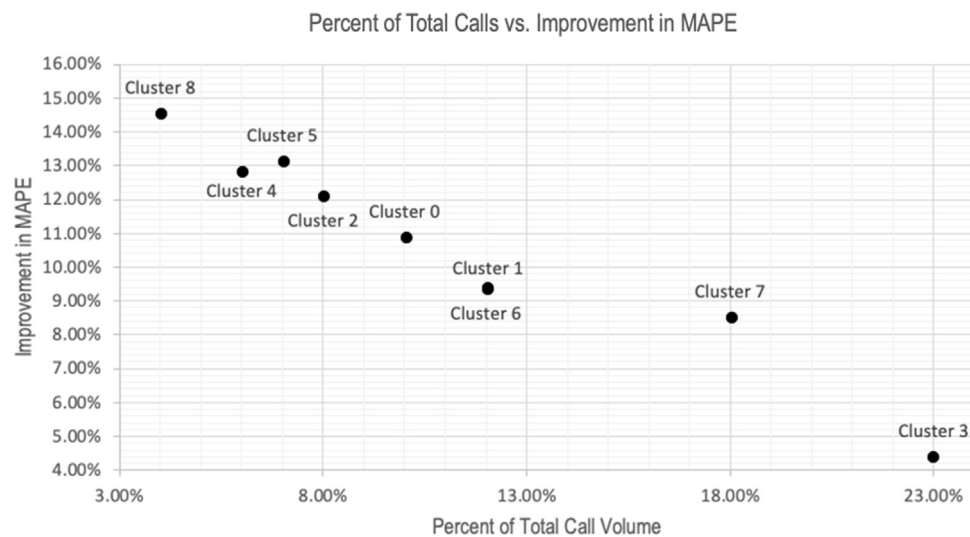


Fig. E.3. Percent of total call volume versus improvement in MAPE values (MLP vs. MHF) by cluster.

Table F.1

MLP models with lagged weather variable results comparison.

MLP model configuration	MAPE	MAD
Daily	5.98	14.54
Daily w/Lagged weather variables	6.21	15.16
Hourly Non-Spatial	27.31	2.79
Hourly Non-Spatial w/Lagged weather variables	27.92	2.70
Hourly 7 Clusters	39.45	0.99
Hourly 7 Clusters w/Lagged weather variables	42.10	0.95
Hourly 8 Clusters	38.11	0.89
Hourly 8 Clusters w/Lagged weather variables	39.06	0.91
Hourly 9 Clusters	35.56	0.82
Hourly 9 Clusters w/Lagged weather variables	38.51	0.86

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